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Incidence of hydrocephalus following decompressive craniectomy for ischaemic stroke: A systematic review and meta-analysis

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ABSTRACT

Objective: Decompressive craniectomy (DC) following malignant ischaemic stroke is a potentially life-saving procedure. Event rates of ventriculomegaly following DC performed in this setting remain poorly defined. Accordingly, we performed a systematic review to determine the incidence of hydrocephalus and the need for cerebrospinal fluid (CSF) diversion following DC for malignant stroke.
Methods: MEDLINE, EMBASE and Cochrane libraries were searched from database inception to 17 July 2021. Our search strategy consisted of "Decompressive Craniectomy", AND "Ischaemic stroke", AND "Hydrocephalus", along with synonyms. Through screening abstracts and then full texts, studies reporting on rates of ventriculomegaly following DC to treat ischaemic stroke were included for analysis. Event rates were calculated for both of these outcomes. A risk of bias assessment was performed to determine the quality of the included studies.
Results: From an initial 1117 articles, 12 were included following full-text screening. All were of retrospective design. The 12 included studies reported on 677 patients, with the proportion experiencing hydrocephalus/ventriculomegaly being 0.38 (95% CI: 0.24, 0.53). Ten studies incorporating 523 patients provided data on the need for permanent CSF diversion, with 0.10 (95% CI: 0.07, 0.13) requiring a shunt. The included studies were overall of high methodological quality and rigour.
Conclusion: Though hydrocephalus is relatively common following DC in this clinical setting, only a minority of patients are deemed to require permanent CSF diversion. Clinicians should be aware of the incidence of this complication and counsel patients and families appropriately.

1. Introduction

Ischaemic stroke is a leading contributor to worldwide morbidity and mortality [1]. Life-threatening cerebral edema with mass effect following middle cerebral artery ischaemic stroke is seen in up to 10% of cases and is termed a malignant infarction [2]. DC in the setting of malignant infarction demonstrates an evident mortality benefit over medical management alone [3–6]. Common complications following this procedure include wound infection and bleeding [3]. Although ventriculomegaly is recognised, the event rate for hydrocephalus requiring CSF diversion following DC for ischaemic stroke is unclear.

Diagnosing clinically relevant hydrocephalus in this patient population is challenging. Poor baseline neurological status due to stroke can hinder attempts at determining symptoms or signs secondary to hydrocephalus. Definitions of ventriculomegaly are confounded by the parenchymal volume loss commonly seen in large vessel ischaemic strokes, and direct measurements of intracranial pressure are difficult to interpret in a patient with a large craniectomy [7]. Ventriculomegaly is extremely common, with some series reporting it occurs in more than half of patients following procedures [8,9], but clinically significant ventriculomegaly is much more difficult to quantify.

Of course, not every patient with ventricular dilatation requires CSF diversion. A more clinically relevant measure may be the proportion of patients requiring surgical intervention for CSF diversion. There is a

Abbreviations: CSF, cerebrospinal fluid; CT, computed tomography; DC, decompressive craniectomy.

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significant discrepancy in the literature regarding the proportion of patients who require permanent CSF diversion following DC to treat ischaemic stroke. Waziri and colleagues were among the first to examine this question. They found that two of 11 (18%) patients undergoing DC for treatment of malignant infarction required permanent CSF diversion via a ventriculoperitoneal shunt [7]. In contrast, a subsequent study identified no patients that required permanent CSF diversion in a similar population of patients [10]. A slightly larger study published in 2013 reported that rates of hydrocephalus requiring CSF diversion approached 10% [11].

Although DC is a life-saving procedure, hydrocephalus requiring CSF diversion is a recognised complication of decompression for malignant infarction. Event rates of hydrocephalus following DC performed in this setting still need to be better defined, with no previous systematic review examining this question. A more accurate estimation of the risk of the need for CSF diversion following this procedure would enhance informed consent discussions with patients and families being considered for this procedure.

2. Materials and methods

A protocol was formulated before the searches for this review were performed that outlined the review question, search strategy, inclusion and exclusion criteria, risk of bias assessment of included studies and the statistical analyses that would be conducted. This protocol was registered with PROSPERO (reference number CRD42021228987) and adhered to the structure dictated by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [12] statement along with the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) checklist [13].

2.1. Definition of hydrocephalus

Varying definitions of hydrocephalus and the overlap with ventriculomegaly highlight the challenge of performing a meta-analysis on this topic. Some studies defined hydrocephalus by combining serial radiological ventricular morphological changes and deteriorating patient's clinical condition [8–10,14]. Other studies solely relied on radiological changes with ventricular dilatation with Evans index > 0.3 and narrowed CSF spaces on CT as meeting the criterion for hydrocephalus [11,15,16]. The remaining studies defined hydrocephalus as elevation in intracranial pressure refractory to medical treatment [7,17,18]. Despite the ambiguity in distinguishing between hydrocephalus and ventriculomegaly, all the included studies recognised persistent ventricular dilation with worsening clinical neurological expression and impaired CSF dynamics as an indication for shunt placement.

2.2. Study selection and search strategy

Study inclusion was determined through the use of a research question formulated in the P (Population) I (Intervention) C (Comparator) O (Outcome) style. The population was patients with large ischaemic strokes causing significant mass effects. The intervention was DC. Any variation on DC will be accepted (unilateral or bilateral fronto-temporo-parietal, bifrontal). The outcome was a requirement for permanent CSF diversion due to hydrocephalus (for example, via a ventriculoperitoneal shunt). There was no comparator for this study. PubMed, EMBASE and the Cochrane Library databases were searched from inception to 17 July 2021. Reference lists of identified articles were screened to identify any relevant publications. No restriction was placed on the time of study publication. No restriction was placed on the language study being published. Studies were excluded if they examined rates of hydrocephalus following DC for indications rather than ischaemic stroke.

The study types included randomised controlled trials and prospective or retrospective cohort studies. The search strategy focused on the following terms: Hydrocephalus OR Ventriculoperitoneal Shunt AND Decompressive Craniectomy AND Ischaemic Stroke OR Malignant Stroke OR Malignant Infarction. Specifically for PubMed (incorporating MEDLINE), search terminology was ((craniectomy* OR shunt* OR diversion*) AND (hydrocephalus OR ventriculomegaly)) AND (stroke OR cerebrovascular OR ischemi* OR ischaemia* OR infarct*). For EMBASE, the search terms were Embase (hydrocephalus OR ventriculomegaly) AND (craniectomy OR shunt OR diversion) AND 'cerebrovascular accident'. For the Cochrane Library, search terminology was ((craniectom* OR shunt* OR diversion*) AND (hydrocephalus OR ventriculomegaly)) AND (stroke OR cerebrovascular OR ischemi* OR ischaemi* OR infarct*).

2.3. Data collection and extraction

Two reviewers (N. A. and K. N.) independently screened identified studies based on title and abstract. Studies included at this step proceeded to full-text review by the same two authors (independently). Disagreements at each stage were resolved by consensus with the aid of a third reviewer (C. O.). The web-based application Rayyan was used to assist the process of abstract/title and full-text screening [19]. Data from included studies were extracted into a prospectively specified form, with extracted data including authors, year of publication, country of origin, number of patients,

average age, side of stroke (right/left), occluded vessel, time to surgery, number developing ventriculomegaly (pre-cranioplasty and post cranioplasty), time to cranioplasty, whether ventriculomegaly improved post cranioplasty, the number requiring CSF diversion procedure, percentage with haemorrhagic transformation or infection as complication and mean follow up duration. The primary outcome was the requirement for permanent CSF diversion to treat hydrocephalus. The secondary outcome was the development of ventriculomegaly/hydrocephalus either defined by the presence of enlarging ventricles on serial radiographic measurements, with an Evans ratio of 0.3 and without the presence of enlarged subarachnoid spaces over the cerebral convexities, or via clinical interpretation.

2.4. Data analysis and risk of bias assessment

Data analyses were performed using Stata Statistical Software (Release 15.1 College Station, TX: StataCorp LP). Proportions of patients with ventriculomegaly following DC and proportions requiring permanent CSF diversion were calculated using a random effects meta-analysis. 95% confidence intervals were calculated. The presence of heterogeneity in the results was tested for through the use of the I^2 statistic (threshold value of >50% considered significant heterogeneity) and the use of Cochran's Q test (p-value of <0.05 regarded as significant heterogeneity). The presence of publication bias was assessed visually through a funnel plot, and a Begg's and Egger's test was performed to determine the existence of slight study bias ($p < 0.05$ considered significant).

Two reviewers (N. A. and K. N.) independently performed the risk of bias assessment of non-randomised studies of health care interventions using the previously validated Downs and Black checklist [20]. This checklist is used to appraise the methodological quality of a research study by considering factors such as the level of data reporting, external validity, bias, confounding and power [20]. The highest possible score is 32, and scores of > 25 were considered excellent, 20–25 as good, 15–19 as fair and < 15 as poor. Discrepancies in scores were resolved by consensus with the aid of a third reviewer (C. O.). Our search did not identify any relevant randomised controlled trials. Still, if present, the risk of bias would have been assessed using the Cochrane Collaboration's tool for assessing the risk of bias in randomised trials [21].

3. Results

3.1. Study selection

Initial searches identified 1206 studies (857 from Medline, 299 from EMBASE and 50 from the Cochrane Library), with 89 duplicates. After removing duplicates, 1117 searches remained for the title and abstract screening. After this step, 1065 studies were excluded, leaving 52 remaining for full-text review. Full-text records could not be retrieved for 9 studies [22–30]. The remaining 43 texts were reviewed, with 11 excluded due to discussing the wrong population, one excluded for reporting on the bad outcome, 17 excluded for reporting on the wrong study type, and two excluded as relevant data could not be extracted. This left 12 studies remaining, which were included. No further pertinent studies were identified from screening reference lists of included studies. The search process is detailed below in Fig. 1 (Base image sourced from Page and colleagues [31]).

3.2. Study characteristics

Characteristics of the 12 included studies are outlined in Table 1 [7–11,14,15,17,32,18,33,16]. All 12 studies were retrospective cohort studies, and the continent of origin was Asia in six, North America in three and Europe in three. A total of 677 patients were included, with 417 males (62%). The mean patient age was 55.8 years, and the mean time craniectomy was performed post-ictus for each study was 2.7 days. Patients with MCA occlusions comprised 533 (79%), and those with ICA occlusions comprised 144 (21%). All studies reported on rates of ventriculomegaly, with 30% (257) developing hydrocephalus/ventriculomegaly. Ten studies incorporating 523 patients said that 52 (10%) patients required permanent CSF diversion for hydrocephalus.

3.3. Risk of bias assessment

Overall, study quality was reasonably high, with an average score of 24.1/32 (76%) on the Downs and checklist. Six studies were of excellent quality, and six were of good quality (See Table 2).

3.4. The proportion of ventriculomegaly/hydrocephalus

Using a Forest plot (Fig. 2), the proportions of patients with ventriculomegaly were pooled across 12 studies using a random effects meta-analysis model. Heterogeneity in the study estimates was assessed using the I-squared statistic (96.1%) and Cochran's Q P value (<0.0001), which showed much heterogeneity. However, the degree of heterogeneity has been accounted for because a random effects model was used. The overall proportion of patients with hydrocephalus/ventriculomegaly across the studies is 38% (95% CI: 0.24, 0.53). A Funnel plot (Fig. 3) shows no publication bias (all twelve studies are in the funnel). Egger's test P value was 0.024, and Begg's test P value was 0.244 (continuity corrected), so the decision on the presence of small-study bias was ambiguous.

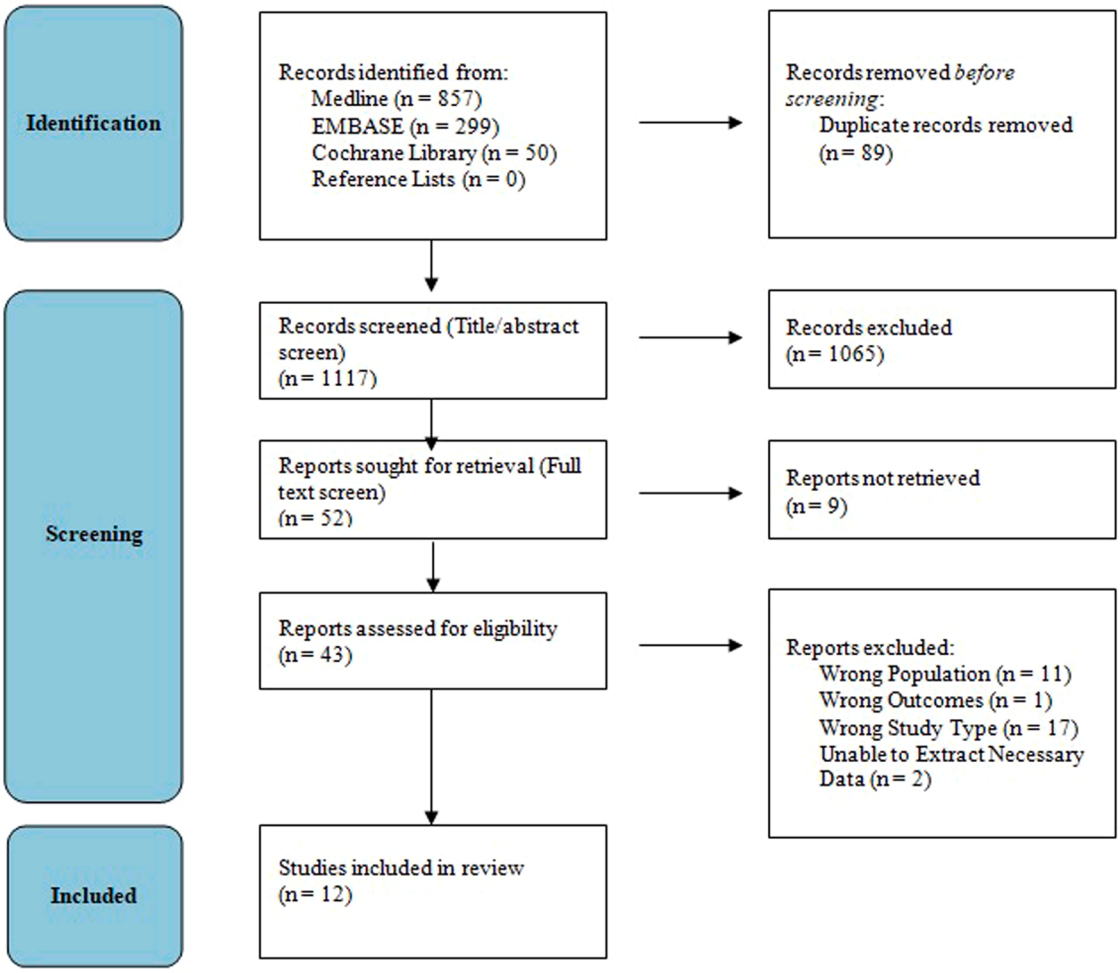

Fig. 1. PRISMA flow diagram detailing how study selection was performed.

Table 1
Characteristics of included studies.

Authors	Year Published	Country	Study design	No. of patients	No. of male patients	Mean patient age
Lin et al.	2015	Taiwan	Retrospective	13	5	62
Finger et al.	2017	Germany	Retrospective	99	61	53
Lee et al.	2012	Taiwan	Retrospective	17	14	62
Nalback et al.	2012	USA	Retrospective	11	3	56
Rahme et al.	2010	Canada	Retrospective	17	8	44
Santana-Cabrera et al.	2016	Spain	Retrospective	23	15	55
Schwake et al.	2019	Germany	Retrospective	198	124	55
Shao et al.	2013	China	Retrospective	131	87	57
Takeuchi et al.	2013	Japan	Retrospective	28	14	64
Takeuchi et al.	2013	Japan	Retrospective	23	13	61
Wang et al.	2016	China	Retrospective	128	67	53
Waziri et al.	2007	USA	Retrospective	11	6	48

Authors	Time to craniectomy post infarct (days)	Involved vessel	Presence of hydrocephalus	Pre cranioplasty hydrocephalus	Post cranioplasty hydrocephalus
Lin et al.	1.8	MCA: 7 ICA: 6	13	13	N/A
Finger et al.	N/A	MCA: 99	10	2	8
Lee et al.	2.6	MCA: 17	7	7	0
Nalback et al.	3	MCA: 8 ICA: 3	1	1	0
Rahme et al.	N/A	MCA: 17	2	2	0
Santana-Cabrera et al.	2	MCA: 23	8	8	N/A
Schwake et al.	1.2	MCA: 140 ICA: 57	15	N/A	N/A
Shao et al.	2.8	MCA: 131	39	39	N/A
Takeuchi et al.	N/A	MCA: 16 ICA: 12	13	13	9
Takeuchi et al.	2.4	MCA: 12 ICA: 11	11	11	7
Wang et al.	3.6	MCA: 78 ICA: 50	73	73	N/A
Waziri et al.	5	MCA: 7 ICA: 4	9	9	3

Authors	No. improved post cranioplasty	No. requiring permanent CSF diversion	Mean time to cranioplasty (days)	Haemorrhagic transformation	Infection	Mean follow up duration (months)
Lin et al.	2	4	75	N/A	N/A	N/A
Finger et al.	N/A	10	108	N/A	N/A	N/A
Lee et al.	2	5	100	5	2	N/A
Nalback et al.	0	0	N/A	N/A	N/A	N/A
Rahme et al.	2	0	21	2	N/A	24
Santana-Cabrera et al.	N/A	N/A	N/A	15	N/A	N/A
Schwake et al.	N/A	15	N/A	N/A	14	N/A
Shao et al.	N/A	N/A	N/A	10	24	N/A
Takeuchi et al.	N/A	2	N/A	N/A	2	5
Takeuchi et al.	N/A	1	57	N/A	N/A	6
Wang et al.	N/A	19	N/A	46	N/A	24
Waziri et al.	2	2	63	N/A	1	N/A

Table 2
Risk of bias scores as assessed using Downs and Black instrument.

Authors	Average Score	Percentage Score
Bon-Jour Lin et al.	25	78
Finger et al.	24	75
Lee et al.	24	75
Nalbach et al.	21.5	67
Rahme et al.	20	63
Santana-Cabrera et al.	26.5	83
Schwake et al.	26	81
Shao et al.	25.5	80
Takeuchi et al.	25.5	80
Takeuchi et al.	20	63
Wang et al.	26	81
Waziri et al.	25.5	80

3.5. Proportions of hydrocephalus requiring permanent CSF diversion

Using a Forest plot (Fig. 4), the proportions of patients requiring CSF diversion were pooled across ten studies using a random effects meta-analysis model. Heterogeneity in the study estimates was assessed using the I-squared statistic (28.8%) and Cochran's Q P value (0.18), which showed minimal heterogeneity. The overall proportion of patients requiring CSF diversion through shunting across the studies was 10% (95% CI: 0.07, 0.13). A Funnel plot (Fig. 5) shows no publication bias (all ten studies are in the funnel). Egger's test P value was 0.145, and Begg's was 0.152 (continuity corrected), so small-study bias was not indicated. Six studies had information on the timing of cranioplasty post-craniectomy, with a reported mean of 71 (21−108) days. No studies delineated the temporal relationship between cranioplasty and improvement in hydrocephalus.

4. Discussion

This study examined the occurrence of hydrocephalus after DC for malignant cerebral infarction and found that 38% (95% CI: 0.24, 0.53) of patients develop this condition, with a 10% (95% CI: 0.07, 0.13) subset deemed to require permanent CSF diversion. This research is significant because it precisely quantifies the need for CSF diversion following DC, which is a more clinically relevant outcome than ventriculomegaly alone, given the variability in definitions across studies.

Before this study, no systematic review had assessed the need for permanent CSF diversion in cases of hydrocephalus following DC for malignant stroke. Earlier reviews on DC for traumatic brain injury reported hydrocephalus rates of 17.9% [34]. Still, their definition relied solely on radiological evidence of ventricular dilatation, leading to a wide range of reported rates (6–54%) [35,36]. Similarly, other studies

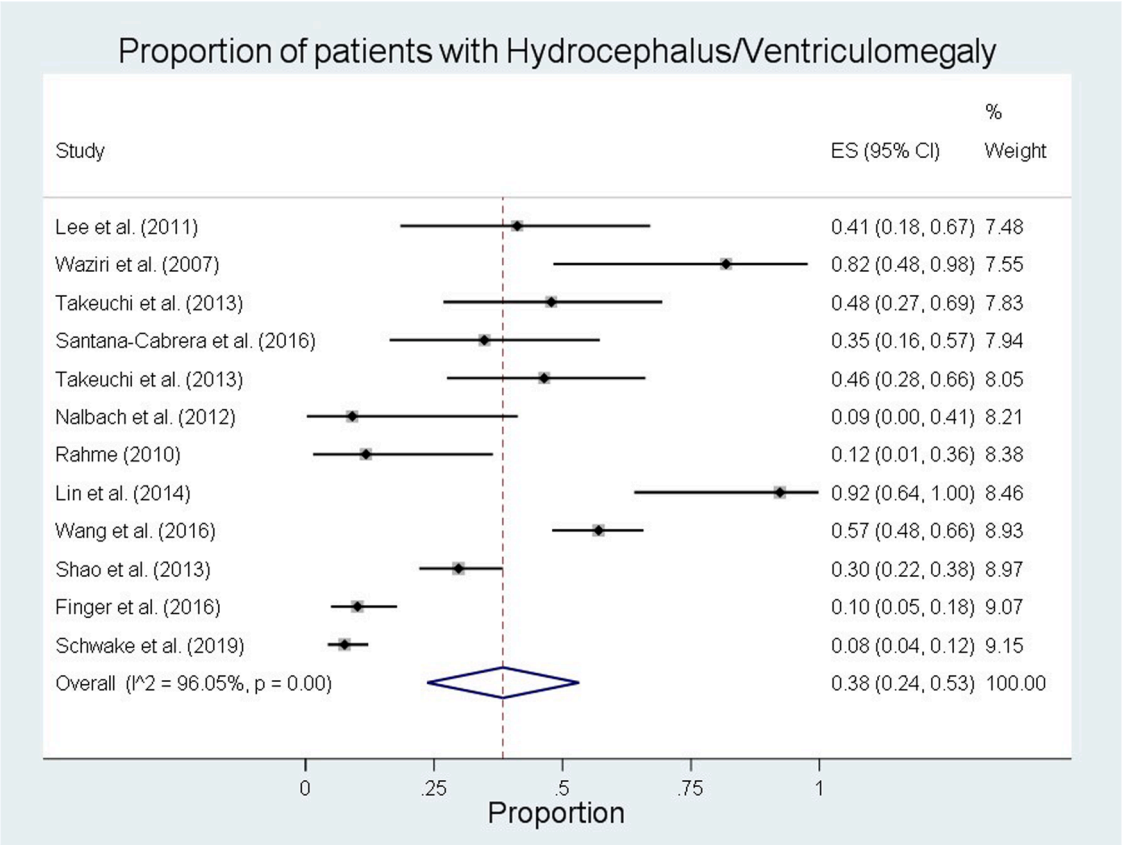

Fig. 2. Forest plot for proportion of patients with Hydrocephalus/Ventriculomegaly.

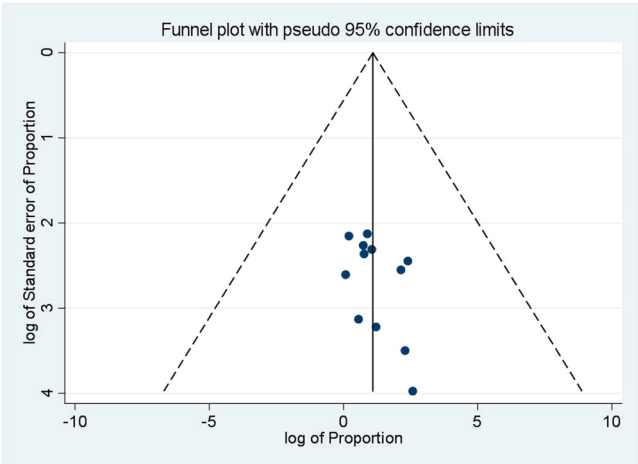

Fig. 3. Funnel plot for proportion of patients with Hydrocephalus/Ventriculomegaly.

reported highly variable rates, ranging from 8% to 92% [8,18], which can be attributed to differing definitions of hydrocephalus and ventriculomegaly.

The discrepancies in reported rates stem from variations in how studies define hydrocephalus and ventriculomegaly. Patients may exhibit ventricular dilatation due to factors other than hydrocephalus, such as brain atrophy [37]. Some studies described hydrocephalus based solely on progressive ventricular dilation on CT scans, leading to high reported rates [7,15,38]. Others had stricter criteria, requiring radiological evidence, clinical deterioration, or a lack of neurological improvement [10].

The wide array of definitions impairs the ability to generalise findings. In contrast, the decision to perform permanent CSF diversion, typically through ventriculoperitoneal shunt placement, is based on clinical and radiological assessments and is a more easily measured outcome.

This study has limitations, as it didn't investigate factors associated with increased hydrocephalus risk, which may have affected the results. Factors like the extent of craniectomy [39] and underlying bridging vein decompression have been linked to hydrocephalus [39]. Additionally, the retrospective nature of most studies and relatively small sample sizes sometimes limit the available evidence.

The study's strength lies in the methodological quality of included studies. However, they were all retrospective, and definitions of hydrocephalus varied, contributing to statistical heterogeneity. Nevertheless, focusing on the clinically relevant outcome of permanent CSF diversion is a strength. Furthermore, there was no significant publication bias, and only minor bias evidence was found for the proportion of patients developing ventriculomegaly/hydrocephalus.

The study highlights the need for further research to understand the efficacy of CSF diversion in managing persistent hydrocephalus after DC and its impact on functional clinical outcomes, ultimately guiding the management of DC-related hydrocephalus.

5. Conclusion

This study is the first systematic review to assess the incidence of hydrocephalus and permanent CSF diversion after DC for malignant cerebral infarction. Approximately 40% experience hydrocephalus, but only a minority have been deemed to require operative methods to divert CSF permanently. This information aids informed discussions with patients and emphasises the need for uniformity in the definition of hydrocephalus and the lack of a standard decision-making tool for

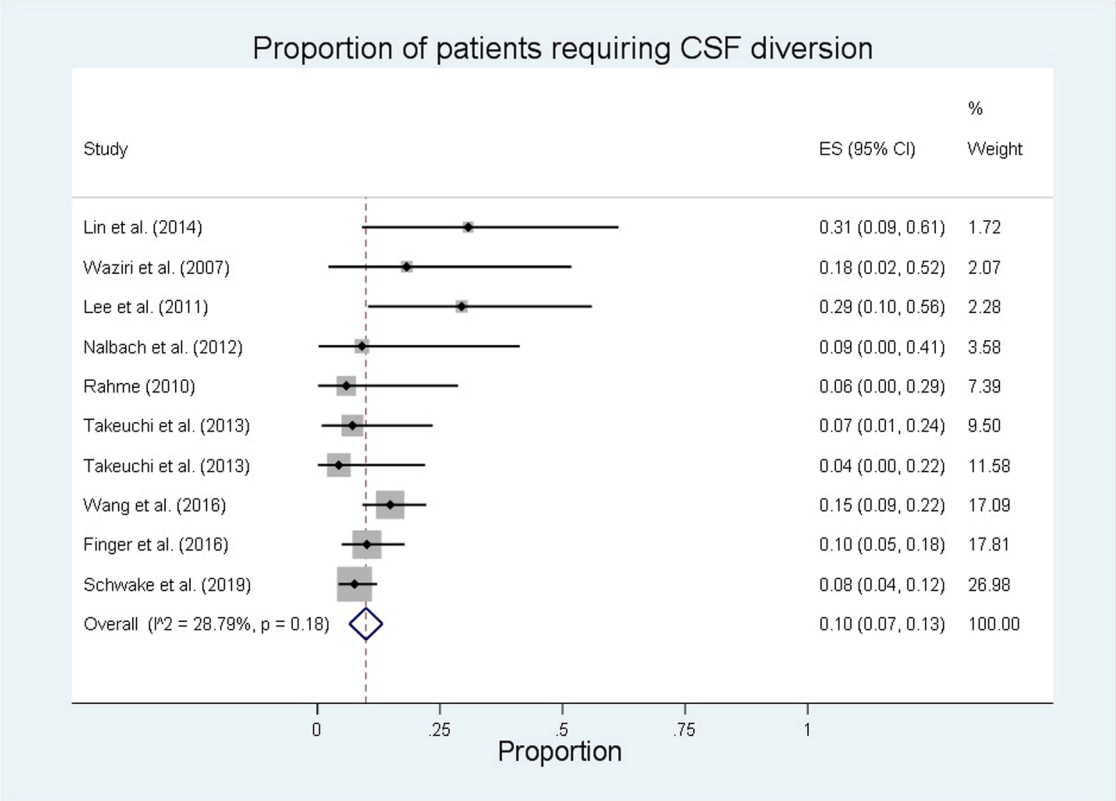

Fig. 4. Forest plot for proportion of patients requiring CSF diversion.

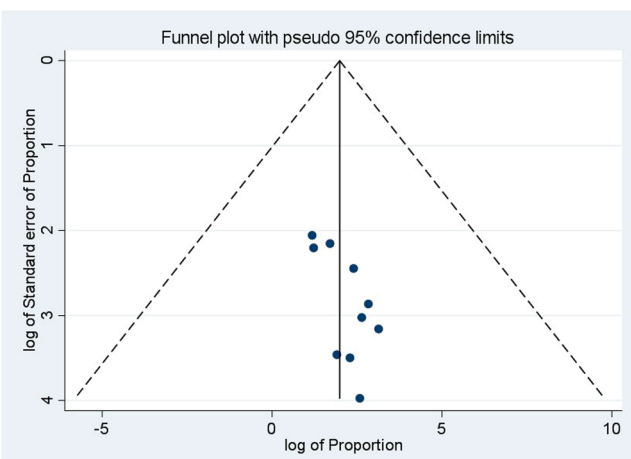

Fig. 5. Funnel plot for proportion of patients requiring CSF diversion.

determining which patients require CSF diversion.

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CRediT authorship contribution statement

Christopher Dillon Ovenden: Conceptualization, Investigation, Methodology, Writing - original draft. **Dwarkesh Dharmendra Barot:** Investigation, Writing - review & editing. **Aashray Gupta:** Investigation, Project administration, Writing - review & editing. **Nidhi Aujayeb:** Resources, Investigation. **Kayla Nathin:** Resources, Investigation. **Joseph Hewitt:** Investigation, Writing - review & editing. **Joshua Kovoor:** Investigation, Writing - review & editing. **Brandon Stretton:** Resources, Writing - review & editing. **Stephen Bacchi:** Formal analysis, Writing - review & editing. **Suzanne Edwards:** Formal analysis, Writing - review & editing. **Lola Kaukas:** Formal analysis, Writing - review & editing. **Adam J. Wells:** Supervision, Writing - review & editing.

Declaration of Competing Interest

None.

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